Project Name

Smart College Resource Waste Detection & Analytics System

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WORKBOOK TEMPLETE

Project Name:

SMART-RESOURCE

Smart College Resource Waste Detection & Analytics System

Problem Statement:

Many colleges use important resources such as electricity, water, food, paper, and internet services every day. However, resource usage and wastage are often recorded separately or not analyzed regularly. The proposed SMART-RESOURCE system collects resource-usage data, identifies waste patterns, compares usage across locations and time periods, and provides analytics to help the college reduce unnecessary consumption and improve resource management.

Why does this problem exist?

• Resource usage may not be monitored regularly or in one centralized system.

• Electricity, water, food, paper, and internet usage can vary significantly by location and time.

• Excessive or unnecessary consumption may remain unnoticed for long periods.

• Manual analysis makes it difficult to identify recurring waste patterns.

• College management may lack clear data-based information for resource planning.

• Without continuous monitoring, it is difficult to measure whether conservation efforts are working.

Who are the stakeholders?

Internal Stakeholders:

👨 🏫 Teachers – support responsible resource use and report resource-related issues.

👨 💼 College Administrator – manages resource records and monitors usage reports.

🏫 College Management – uses analytics to make resource-saving decisions.

💻 System Administrator/Developer – maintains the application, database, and analytics dashboard.

External Stakeholders:

👨 🎓 Students – use college resources and can participate in resource-saving activities.

👪 Parents/Guardians – benefit indirectly from efficient use of college resources.

🏢 Service Providers/Vendors – may use relevant consumption information for planning and supply management.

Expected system users:

Students – view awareness information and resource-saving recommendations.

Teachers – report resource issues and view relevant usage summaries.

Administrator – manage resource categories, locations, records, and reports.

Management – view overall consumption, waste patterns, trends, and recommendations.

Objective of the project:

To develop a smart resource-waste detection and analytics system for a college.

To collect and maintain organized data on electricity, water, food, paper, and other selected resources.

To identify high-consumption areas, unusual usage, and recurring waste patterns.

To compare resource usage by department, location, day, month, or other suitable periods.

To provide data-driven recommendations for reducing unnecessary resource consumption.

To provide useful dashboards and reports for administrators and college management.

To track resource-saving improvements over time.

Expected outputs:

Display resource consumption records and usage summaries.

Show resource-wise, location-wise, and time-wise consumption trends.

Identify areas or periods with unusually high usage or possible wastage.

Generate dashboards with charts, key performance indicators, and comparisons.

Provide resource-saving recommendations based on observed patterns.

Track changes in consumption before and after improvement measures.

Generate periodic resource usage and waste-analysis reports.

Maintain accurate and updated resource records.

Hands-on-Activity:

Conducted a survey to understand common resource-usage and wastage problems in a college environment.

Collected sample data such as electricity usage, water usage, canteen food sales and leftovers, paper usage, or internet usage, depending on availability.

Prepared a data-gathering table with required fields, units, dates, locations, and data types.

Designed a basic website/application interface for recording and monitoring resource usage.

Created a resource analytics dashboard showing consumption, trends, comparisons, and possible waste areas.

Prepared a basic workflow from data collection to cleaning, analysis, waste detection, recommendation, and reporting.

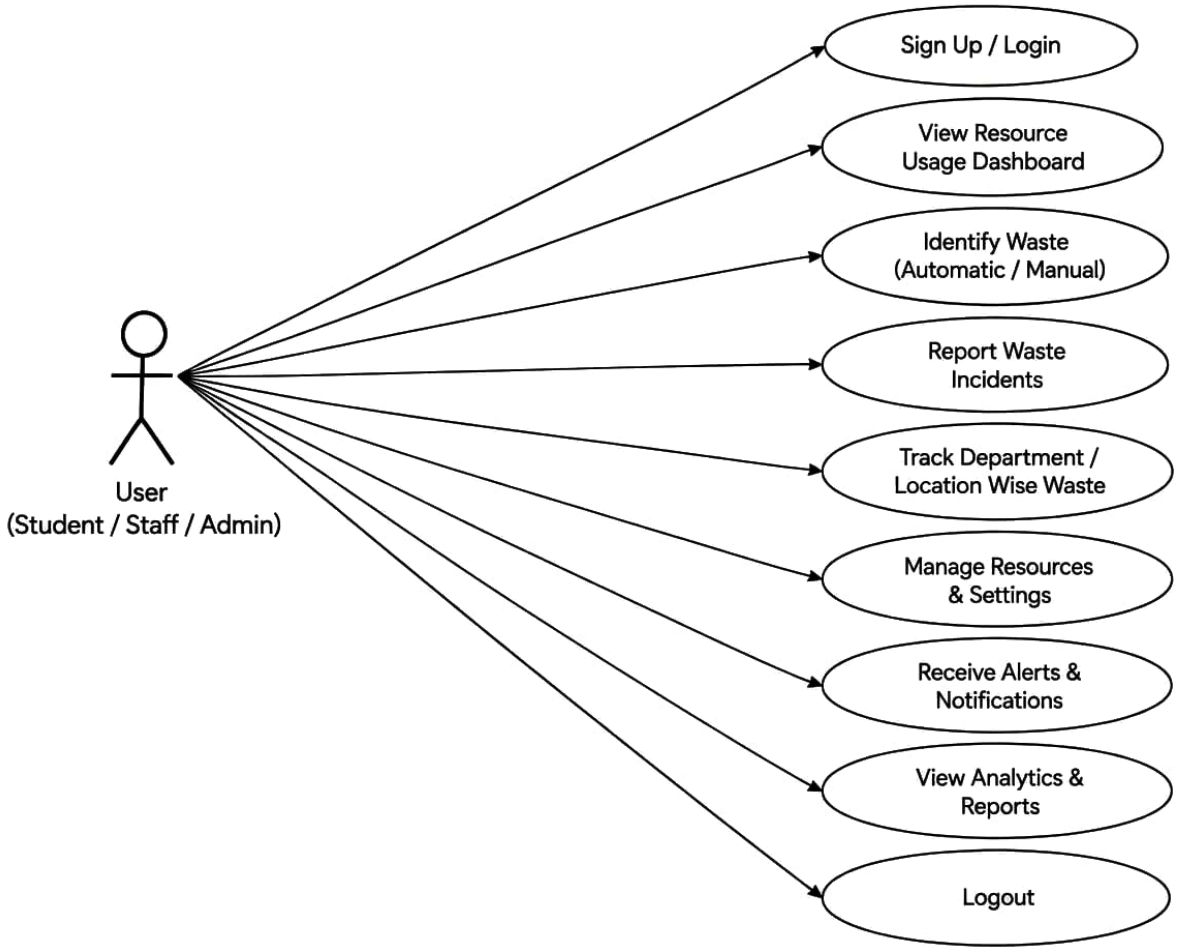
Identified stakeholders and expected system users.

Discussed how data analytics and suitable machine-learning techniques can help detect unusual

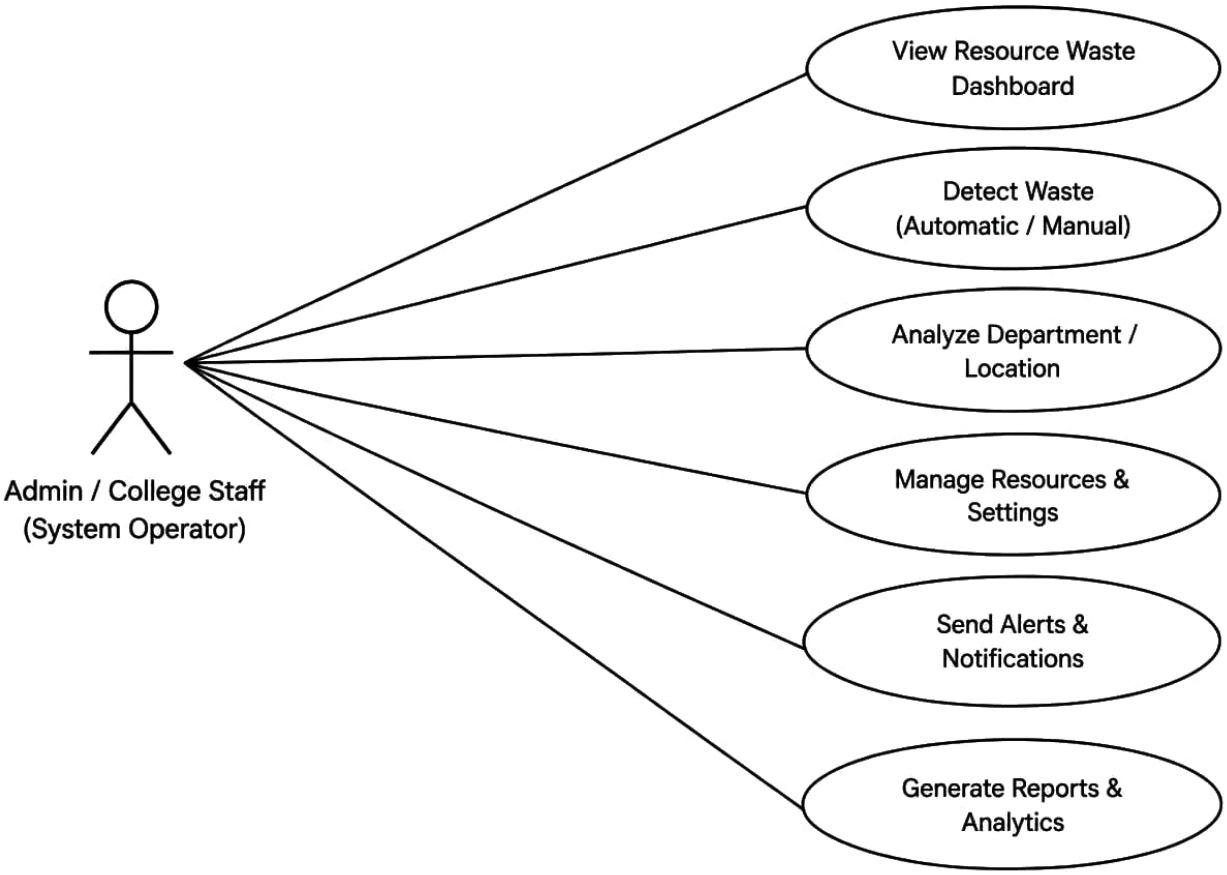
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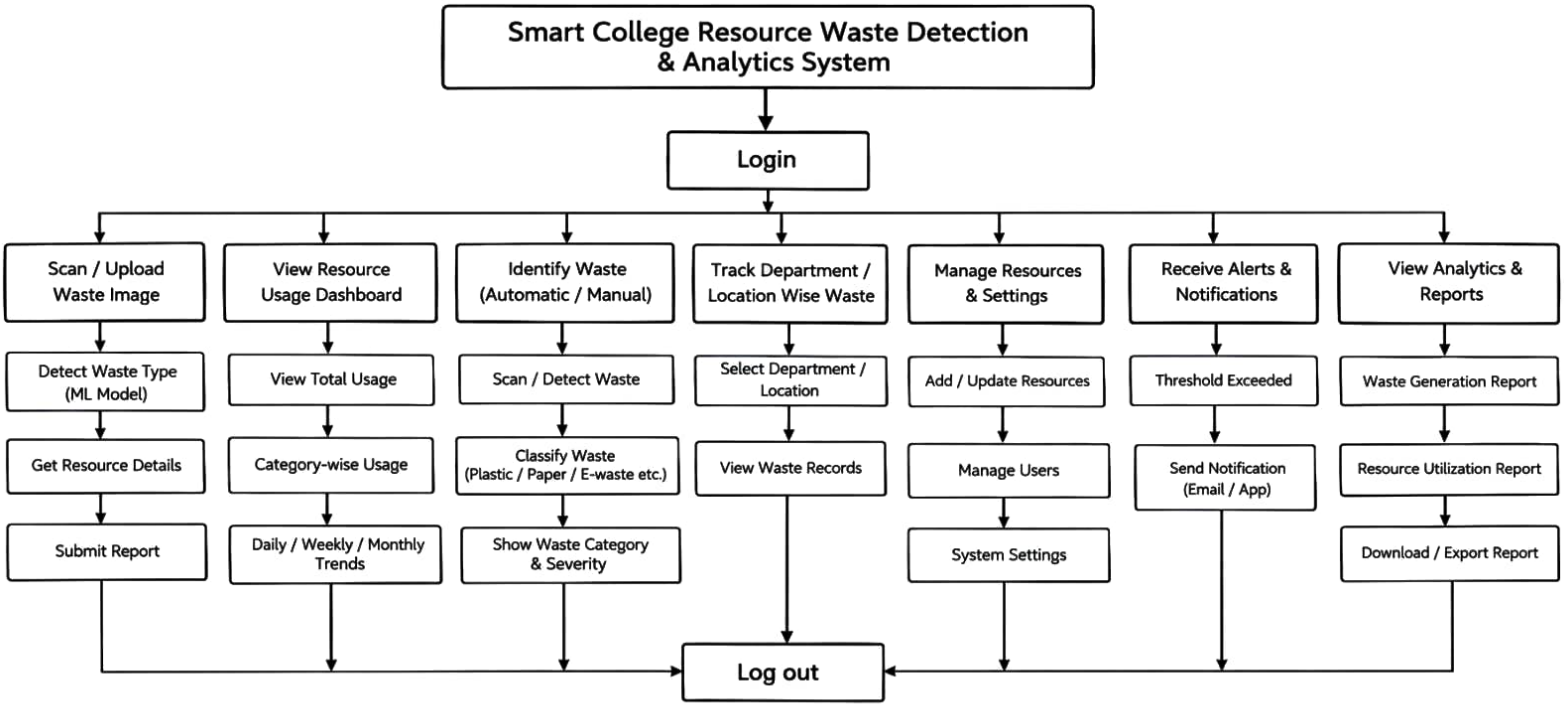
consumption patterns and support better resource management.

Smart College Resource Waste Detection & Analytics System



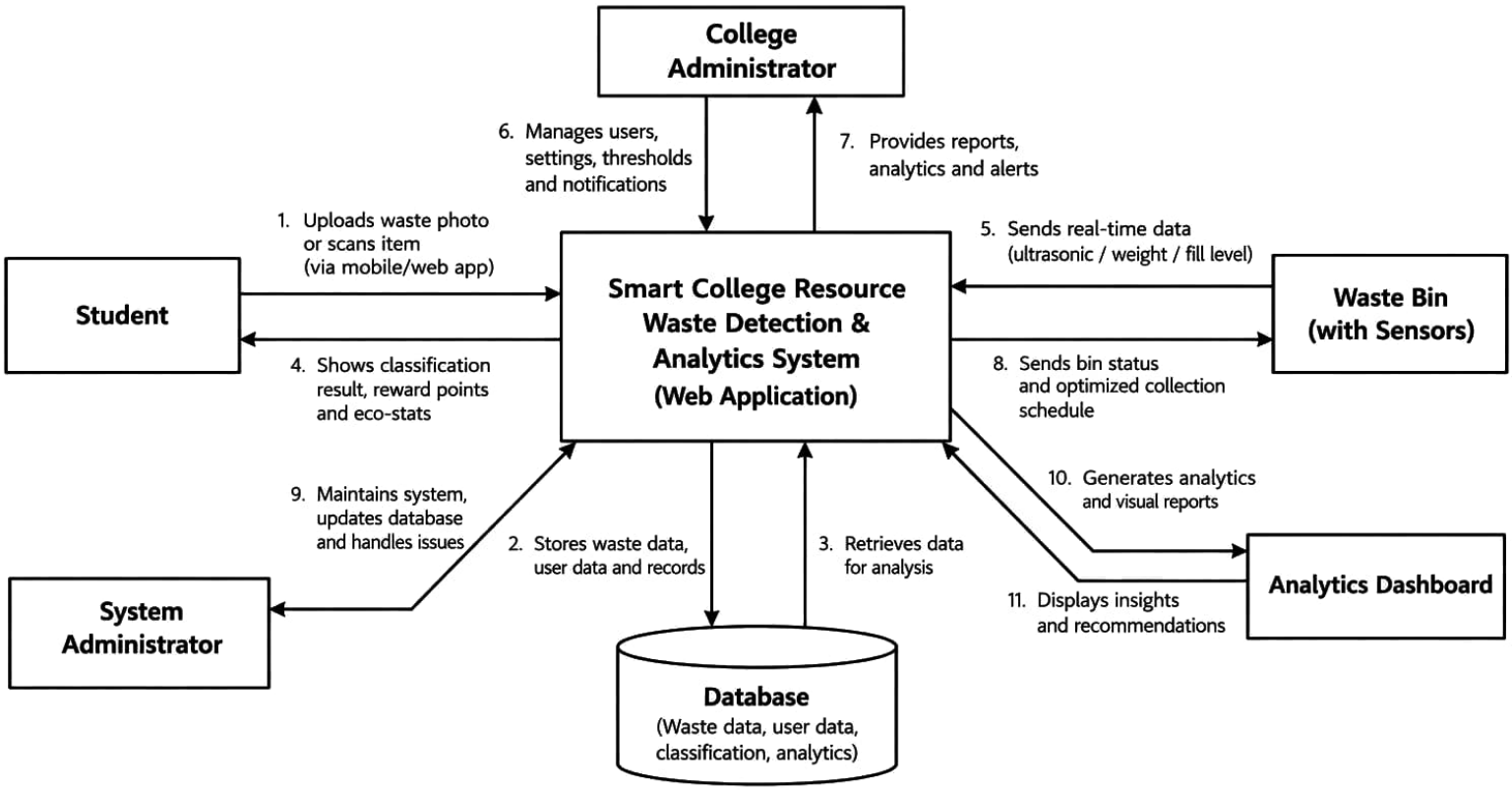
Smart College Resource Waste Detection & Analytics System




Smart College Resource Waste Detection & Analytics System
Login
Scan / Upload Waste Image
Detect Waste Type (ML Model)
Get Resource Details
Submit Report
View Resource Usage Dashboard
View Total Usage
Category-wise Usage
Daily / Weekly / Monthly Trends
Identify Waste (Automatic / Manual)
Scan / Detect Waste
Classify Waste (Plastic / Paper / E-waste etc.)
Show Waste Category & Severity
Track Department / Location Wise Waste
Select Department / Location
View Waste Records
Manage Resources & Settings
Add / Update Resources
Manage Users
System Settings
Receive Alerts & Notifications
Threshold Exceeded
Send Notification (Email / App)
View Analytics & Reports
Waste Generation Report
Resource Utilization Report
Download / Export Report
Log out

Smart College Resource Waste Detection & Analytics System

(Collaboration Diagram)



Smart College Resource Waste Detection & Analytics System
(Class Diagram)

